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ORIGINAL RESEARCH

Deep Semantic Segmentation Assisted Region-of-Interest Sensitive Deep Spatio-Textural Feature Learning Framework for Leprosy Detection and Classification

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Abstract

The last few decades have witnessed exponential rise in leprosy also called Hansen diseases globally. Being chronic and infectious in nature, eradicating leprosy has remained a challenge. Despite a few recent efforts employing vision computing-based leprosy detection and classification, most of the at hand solution remain confined due to their inability to address data-imbalance, feature optimality, reliability and scalability. Addressing aforesaid challenges requires a vision-based method to accommodate automated region-of-interest (ROI)-specific information-rich learning and classification. In this paper, a highly robust deep semantic segmentation assisted region-of-interest sensitive deep spatio-textural feature learning framework is proposed for leprosy detection and classification. The proposed method at first applies the different pre-processing methods including contrast adaptive histogram equalization, intensity equalization, resizing. In sync with probable data-imbalance problem, we applied deep semantic segmentation model to obtain the ROI specific regions, which were subsequently processed for RGB conversion. Thus, the obtained ROI-specific RGB images were processed for feature extraction by using deep spatio-textural feature extraction model encompassing GLCM and ResNet101 deep network. Here, GLCM extracted sufficiently large spatio-textural features, while the use of ResNet101 not only provided deep residual features but also alleviated the likelihood of gradient fading problem, which is highly frequent in the state-of-arts like convolutional neural network (CNNs) or other recurrent neural networks (RNNs). Thus, the obtained deep spatio-textual features processed for horizontal concatenation-based feature fusion, which was later trained by using random forest ensemble classifier.The simulation results revealed that the proposed model exhibits accuracy and F-Measure of 97.20% and 97.80%, respectively. Thus, the ability to yield superior accuracy along with the potential to alleviate class-imbalance, feature inferiority and gradient fading makes proposed model more reliable and efficient towards real-time leprosy detection and classification.

Keywords Leprosy detection and classification · Deep spatio-textural feature · GLCM · ResNet · Ensemble learning · CAD

Introduction

The last few years have witnessed high-pace rise in global population and allied healthcare demands. The rising health complications and allied diversity of diseases have always remained challenge for government agencies as well as health industry. Despite non-uniform case distribution, Leprosy, also called Hansen's disease, has become a disease of concern globally. In general, Leprosy disease is caused by Mycobacterium leprae [1] that makes it chronic infectious. Though, it can also be caused at the human body locations, underwent treatment with multi-drug therapy (MDT) [1, 2]. Despite lower frequency, if unrecognized in early period and remained unmanaged, it can also cause irreversible nerve damage and deformities in human body reactions [2].

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Typically, its impact can be found more severe over the human skin, especially in the vicinity of the surrounding eyes, nerves and the mucosal surfaces pertaining to the upper respiratory tract [2]. The fact that the leprosy can take place to an individual of any age and gender indicates its criticality. Despite the fact that the leprosy belongs to the curable disease category, the lack of earlier detection and diagnosis can prolong it to years and hence can not only impact the health conditions of the patients but can also make it infectious [2, 3]. The aforesaid diagnosis delay can also cause varied health adversaries or disabilities that can impact well-being. As stated earlier, being communicable in nature, the delayed detection and diagnosis can make it more infectious due to the droplets from mouth and nose, physical contacts etc. [2, 3]. Exploring recent report by World Health Organization (WHO) indicates that in the recent years the leprosy cases have increased significantly in India, Brazil, Indonesia, Nepal, Congo, Somalia etc. [4]. The allied health challenge can be understood by the fact that the leprosy cases has been found in tropical and subtropical conditions ranging over 149 countries globally. The studies indicate that a large population living in poverty and inadequate sanitation might undergo severe leprosy infection. Unfortunately, India being an economically developing nation contributes almost 57.7% of the global cases in leprosy that triggers the nation to develop more efficient and timely detection and diagnosis (India Leprosy Data, 2018) [5–7]. Despite billions of healthcare investments globally, alleviating or eradicating leprosy has remained a challenge. The key reason behind such challenges can be the lack of earlier detection and diagnosis, complex chemical assay treatments, social stigma and lower social sensitivity etc. [8]. Unlike other skin diseases such as the melanoma, leprosy is found non-linear in structure, thus making overall diagnosis more a challenging and complex [8]. The classical leprosy detection methods are more complex and exhaustive that limits its scalability to meet rising demands [9]. Moreover, it needs skilled healthcare experts to detect leprosy cases which makes it human reliant but exhaustive [10–12]. The existing chemical analysis-based treatment methods are limited and are often found resulting false positive outcomes due to the lack of accurate and delayed detection approaches [11, 12]. The at hand method functional on the basis of the hypothesis that a skin lesion possessing five or fewer skin lesions without any significant bacilli presence on skin smears, can be categorized as the paucibacillary (PB) leprosy. A lesion possessing merely one lesion is referred as a single lesion PB. Those skin lesions possessing six or more lesions possessing a positive skin smear test can be labelled as a multibacillary (MB) leprosy [1–5]. Yet, these method needs very experienced healthcare professional to assess skin lesions and a mistake over assessment might result into wrong diagnosis decisions. The classical diagnosis measures employ assessment of the physical behaviors like reddish skin patches with sensory loss, paresthesia's with allied numbness in extremities, painless burns, lumps or swelling on the earlobes and increased sensitized peripheral nerves; yet it can be insufficient to cope up with real-world demands [5, 6]. Additionally, due to the limitations and scarcity with low lab facilities, WHO recommends leprosy profiling on the basis of the lesions present on skin [12]. To cope up with scalable leprosy detection and classification demands, vision-based methods were proposed. However, it requires manual skin-lesion analysis which is complex and difficult in real-time realization [13]. Unlike classical invasive and chemical analysis methods, vision-based approaches can yield superior and scalable leprosy detection and classification systems [14]. The potential to exploit spatial and textural features from the input skin (leprosy) lesions makes vision-based approaches viable towards leprosy detection and classification [14]. Literatures indicate that the strategic amalgamation of spatio-textural features like non-linear patterns (i.e., variations in color, contrast, dimensions, etc.) can be employed for leprosy detection and classification [14]. In this reference, a few efforts have been made by exhibiting two-dimensional (2D) image analysis and exploiting local spatial and textural features [15]. Despite the suitability to yield scalable leprosy detection and diagnosis solutions, training a machine learning model over the region of interests (ROI) remains challenge. Training any machine learning model over non-ROI specific features might skew the overall efficiency [15]. This is because in real-world lesion images, the ROI can be significantly lower than the non-ROI or background regions, and hence training over the skewed majority class background spatio-textural features can impact overall classification results [14]. It indicates that the vision-based diagnosis models require learning over the ROI-specific spatio-textual features to yield accurate learning and classification. It demands more efficient and robust feature extraction model followed by classification environment to perform leprosy detection and classification [15]. In sync with aforesaid inferences, different local spatio-temporal features were applied towards leprosy detection and classification [15–17]. A few approaches like [16] exploited local textural features including gray-level co-occurrence metrics (GLCM), local binary patterns (LBP), discrete cosine coefficients, wavelet features for leprosy detection in skin-lesion images. Yet, non-linearities in the skin lesions, especially over minimally annotated data makes them vulnerable to undergo false positive performance [16, 17]. The predominant reason behind aforesaid inferiorly exhibiting models can be ambiguous ROI definition and insufficient intrinsic features [15]. Though, to alleviate aforesaid feature limitations deep learning methods like convolutional neural network (CNN) [27] and transferable deep network [18–22] have been employed towards leprosy detection and

classification. However, these methods are found limited due to the low-dimensional local features and the impact on non-ROI features on overall learning [23–26]. The literatures indicate that the use of local spatio-textural features including high-dimensional deep feature learning can make learning effective towards leprosy detection and classification, even with smaller number of annotated data. Additionally, the use of ROI-specific features for learning and classification can help to minimize insignificant feature learning and hence can make detection and classification more accurate [16, 17, 26]. In the past, varied ROI segmentation methods are proposed including clustering, static threshold, region-growing etc.; however, spatial non-linearity makes those approaches limited towards leprosy skin lesions which possesses significant diversity based on lesion presence over the body part(s). Moreover, the use of heuristic driven segmentation can be delaying and computational costly and might require manual centroid definition to perform ROI segmentation. To alleviate it, the use of deep segmentation methods, especially the semantic segmentation can be vital. Thus, the use of semantic segmentation assisted ROI specific feature extraction and learning can yield more automatic, accurate and scalable leprosy detection and classification system.

In this paper a robust deep semantic segmentation assisted region-of-interest sensitive deep spatio-textural feature learning framework for leprosy detection. As the name indicates, the proposed model at first applies the different preprocessing algorithms including contrast adaptive histogram equalization, intensity equalization and resizing that consequently helped alleviating the impact of the local illumination conditions. Subsequently, to ensure ROI-specific feature learning deep semantic segmentation model was applied. More specifically, we applied UNet segmentation to detect ROI and non-ROI regions, where the later was dropped from further computation. Once segmenting the leprosy tumor or lesion specific region(s), to further extract deep spatio-textural features (DSTF), the segmented ROI was processed for multiplication with the original ROI, which yielded the ROI specific regions in RGB color space. It enabled extracting both ROI specific deep as well as spatio-textural features for further learning. The segmented ROI-specific RGB images were subsequently processed for ResNet101 based feature extraction. Noticeably, unlike traditional CNN or equivalent deep networks which are often criticized for their inability to address problems like gradient vanishing which can have decisive impact on the overall learning and allied leprosy prediction results. Such problems can be severe over the input images possessing high spatio-textural feature non-linearity, which seems to be highly frequent in at hand leprosy images. To alleviate this problem, we applied a well-known residual network named ResNet50 to extract ROI-specific features from the input images. Additionally, to enrich features with spatial details, GLCM method was applied. Thus, extracting the GLCM and ResNet101 features, horizontal concatenation was performed, which was followed by Random Forest ensemble-based two-class classification. Here, RF algorithm classified each input image as "Leprosy Detected" or "Normal" and labelled them as 1 and 0, respectively. Applying confusion metrics, the performance analysis was done, where the proposed model exhibited the prediction accuracy of 97.20% and F-Measure of 97.80%. The relative performance too confirmed superiority of the proposed model over other state-of-arts, signifying efficacy towards real-time leprosy detection and classification solution.

The other sections are divided as follows. Section "Related Work" presents the related work. Section "Proposed System" presents the proposed method, while the simulation results are discussed in Section Results and Discussion. The conclusion and inferences are given in Section "Conclusion", while the references used are given at the end of the paper.

Related Work

The literatures indicate that the majority of the state-of-arts towards leprosy detection employ chemical assay and allied histological assessment approaches [20–25]. However, these approaches are resource consuming, delaying and possess number challenges towards scalability [26]. Unlike aforesaid chemical assay methods, the vision-based methods have gained widespread attention towards leprosy detection [17–19, 26]. Vision-based approaches have exploited spatio-textural methods to perform different computer aided diagnosis (CAD) methods to perform melanoma cancer detection, leprosy detection and classification [16, 17]. However, guaranteeing reliability of the solution due to high resilience on spatio-textural features and larger samples confine majority of the state-of-arts. The vision-based methods involve ROI detection, feature extraction and learning towards leprosy detection and classification; however, it requires sufficiently large spatio-textural features pertaining to the ROIs to ensure reliability. In sync with the innovations in vision-technology the different efforts are made for ROI-segmentation, feature extraction and classification. Yet, identifying the optimal set of computing environment over non-linear and minimally annotated data has remained a challenge. Recently, deep learning methods were also applied for leprosy detection. For instance, the authors [27] applied CNN deep network to extract deep feature, which was subsequently used to perform entropy-based classification for leprosy detection. Additionally, the Google inception-v3 deep network was applied to learn deep features to enhance performance; however, it could achieve the highest accuracy of 90% signifying its limitations to ensure reliability. Jain et al. [28] too applied traditional deep learning method for leprosy detection that exhibited the maximum

leprosy detection accuracy of 91%. In sync with spatio-textural feature diversity over leprosy lesions Mondal et al. [29] proposed a lesion-specific learning method to classify input skin lesion(s) as leprosy, Tinea Versicolor, and Vitiligo. They applied Generative Adversarial Network (GAN) to generate the synthetic data, which was subsequently learnt by using DenseNet-121 deep network. This approach yielded the highest accuracy of 94.25%. Despite their claim to have 11% superior accuracy than the features without augmentation [26], its efficacy remains suspicious especially when the annotated data sample is less. It questions its suitability towards at hand leprosy detection and classification task. The authors [26–29] employed deep network with low dimension features. To improve aforesaid approaches, Lo et al. [30] designed deep CNN (DCNN) that in conjunction with AlexNet provided high-dimensional feature set to perform bacillus image block detection and classification. This method exhibited the accuracy, sensitivity and specificity of 95.3%, 93.5% and 96.3%, correspondingly. Though, the same approach without applying color information exhibited the accuracy, sensitivity and specificity of 73.8%, 70.7% and 75.4%, correspondingly. Studies [27–29] revealed that the use of DCNN with AlexNet network can classify bacilli tissues from other tissues. The authors [16] made use of the different features including binary patterns, gray level co-occurrence matrix for leprosy detection. The authors applied support vector machine (SVM) to perform classification where LBP-feature with SVM classifier exhibited superior performance (accuracy 89.65%). The authors [26] stated that the information rich features yield superior learning efficacy to achieve Leprosy detection and classification [31]. The authors [32] used machine learning to detect the subtle changes in myo-electric signals to detect Lesion afflicted components. However, high resilience on manual feature extraction, labelling and training confines their scalability. Pal et al. [33] multiple local binary patterns feature to perform skin lesion classification. The authors [34] used magnetic resonance imaging to examine ulnar nerve abscess in leprosy; yet, its scalability remains suspicious over large inputs with high spatio-textural complexities. Despite efforts [34–41] for skin lesion segmentation, guaranteeing optimality over structural complexity and non-linear lesion patterns [42, 43]. It revitalizes researchers to achieve better leprosy detection and classification model which could yield highly accuracy and reliability [43–46].

Proposed System

The overall proposed model encompasses the following phased implementation.

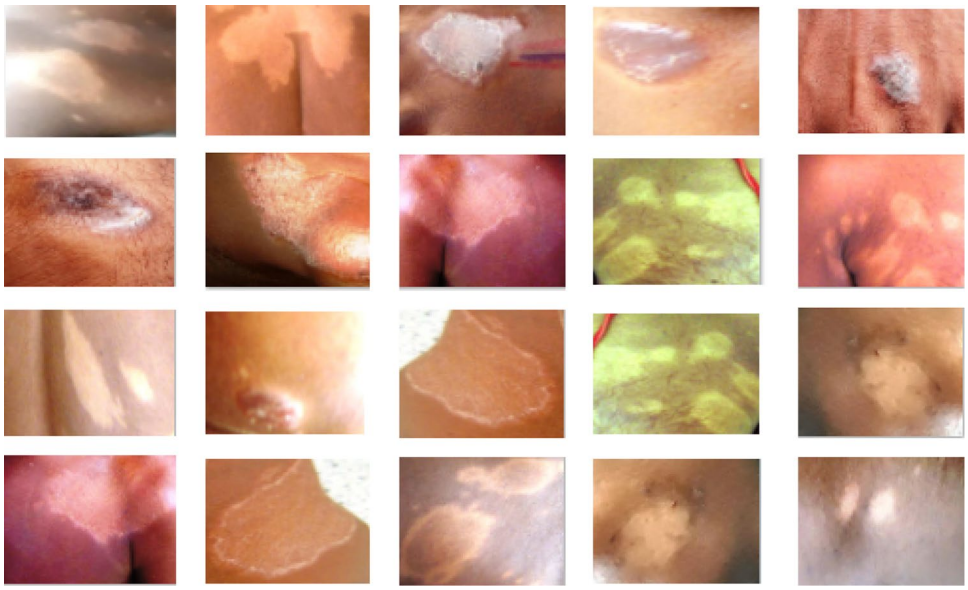
Phase-1 Data Acquisition and Pre-processing
Phase-2 Semantic Segmentation
Phase-3 ROI-Specific Feature Extraction
Phase-4 Feature Fusion, and
Phase-5 Machine Learning Classification.

A. Data Acquisition and Pre-Processing

Despite a few efforts most of the at hand approaches have applied limited numbers of data with no significant spatio-textural diversities. It can limit their efficacy to cope up with scalable system demands. To alleviate this problem, in this work primary data collection was performed, where normal camera-based acquisition was done to collect sufficiently large samples. The primary samples were collected from the different hospitals and leprosy related institutions involved in diagnosis, treatment and rehabilitations. In addition, online repositories providing snippet of the leprosy and allied spatio-textural diversities were also taken into consideration. Recalling the fact that a robust vision-computing model requires input images to have sufficiently large spatio-textural feature diversities including the different color, positions and shapes of the leprosy related skin lesions, a rigorous sampling was performed. The data sampling was done in such manner that the input images (say, training images) possess diversity in organs or the body parts lesions, dimensions or shapes differences as well as color differences. In other words, diversity of features was ensured by selecting skin lesions from the different locations such as thigh, fingers, nose (including nose side), face, limbs, palms, back side images etc. Similarly, the data were selected from the patients of different skin colors, skin and lesion gradients. The initial data sampling encompassed a total of 220 images pertaining to the different leprosy cases. To ensure data suitability and intrinsic feature presentation, the input images were processed for data augmentation and resampling that obtained samples with the different orientations and field of views etc. Thus, a total of 450 leprosy related images were obtained. In reference to the two-class classification problem (Normal Image and Leprosy Detected), we also considered normal skin images as well, which was collected from the different body parts of the patients pertaining to both male as well as female genders. A snippet of the input images considered is given in Table 1.

In sync with spatio-textural feature's diversity over the input samples from the different patients, normalizing the input spatial feature is must. Additionally, addressing clutters, contrast variations and illumination impact is equally significant. To achieve it, different pre-processing methods were taken into consideration. In this work, the different pre-processing methods were taken

Table 1 Illustrations of the leprosy lesion images



into consideration including contrast adaptive intensity equalization, histogram equalization and z-normalization to enable normal pixel distribution across images. For histogram equalization those images in which the frequency distribution representing histogram values were taken into consideration. Once estimating the histogram values of the input samples, the histogram specification and corresponding equalization model was applied to match the histogram values of the different RGB leprosy images (in comparison to the reference images). Here, histogram equalization can enable uniform pixel resolution across the input images that eventually can support efficient feature extraction and learning. The image quality of the images signifying leprosy lesions were later processed for color normalization and contrast adaptive normalization. To minimize the variations of the illumination across input images, this work also made use of the contrast adaptive normalization technique in which the contrast was minimized in such way that the distance of a pixel forms the center of the input image. In sync with the higher non-linearity over input images, contrast adaptive histogram equalization or improvement method was taken into consideration. The proposed model applied Eq. (1) to perform local contrast improvement in each input image.

$$p_n = \left(\frac{\left[u_w(p) - u_w(Min)\right]}{\left[u_w(Max) - u_w(Min)\right]} \right) \tag{1}$$

In (1), p refers the pixels of the input leprosy image(s), while the parameters Min and Max signifies the minimum and the maximum intensity in the moving window. The variable u_w states the intended output's minimum and maximum intensity. Mathematically, it is defined as per the Eq. (2).

$$u_w = \left[1 + exp\left(\frac{f_w - p}{\sigma_w}\right)\right]^{-1} \tag{2}$$

In (2), the parameters f_w and σ_w signifies the mean and standard deviation within the window, which are measured by using Eqs. (3) and (4), correspondingly.

$$f_w = \frac{1}{N^2} \sum_{(i,j)\in w(x,y)} p(i,j) \tag{3}$$

$$\sigma_w = \frac{1}{N^2} \sum_{(i,j)\in w(x,y)} \left(p(i,j) - f_w\right)^2 \tag{4}$$

In (4), the parameter N represents the size of the window, while x and y signifies the pixel location over the moving window. It helped improving the contrast of the input leprosy images. To ensure better ROI segmentation, the input images were processed for resizing by applying Matlab resizing function, which retained the input images in the dimension of [400 × 496] pixels. The input images were converted into equivalent gray scale images by applying RGB2GRAY transformation function. It enabled in retaining gray-level feature representation of the input leprosy images.

Realizing that undeniable feature diversity and non-linearity over the input images, especially over the ambiguous and unclear boundary conditions the proposed model executed z-score normalization. It made use of the statistical details like mean (μ) and standard deviation (σ). This method guaranteed the feature

distributions mean as zero and standard deviation as 1. In this work, we applied Eq. (5) to measure Z-score value (x) for $i-th$ image.

$$x^i = \frac{(x-\mu)}{\sigma} \tag{5}$$

Thus, once performing normalization, the segmentation method was executed, which is discussed as follows:

B. Semantic Segmentation

Majority of the classical segmentation approaches have applied static threshold methods to perform ROI segmentation; however, it can't be suitable over at hand leprosy image segmentation, especially over the non-linear feature spaces. Additionally, the approaches involving clustering methods are found limited due to manual seed point definition and random centroid usages that eventually makes clustering ineffective over exceedingly high non-linear feature space. To alleviate aforesaid challenges and enable efficient segmentation over non-linear input features, semantic segmentation methods can be of great significance. Considering this fact, in this work, a novel fully convolutional network driven UNet architecture is designed for ROI segmentation over input leprosy images. The proposed UNet model was designed in such manner that it could achieve accurate ROI segmentation, even over smaller number of training images. In UNet architecture, the key approach was to accommodate a typical contracting network with successive layers, which can be achieved by substituting the pooling operators by up-sampling operators or functions. This as a result can enhance overall resolution of the images as well as ROI specific area. In order to locate the high-resolution features (in input images), the features from the contracting path were amalgamated with the up-sampled output. Moreover, a convolutional layer was applied to learn the aforesaid amalgamated features to perform more accurate segmentation. To improve efficacy of the proposed UNet segmentation model, the up-sampling component was designed to have a large number of channels, which helped the network to propagate contextual information to the high-resolution layers. In this manner, the expansive path resembles almost symmetric to the contracting path and hence looks like a U-shape architecture. Though, it doesn't employ fully connected layers and applies merely the genuine component of each convolution, signifying the segmentation map comprising the pixels for which the overall context is present within the image. Noticeably, in biomedical image processing, where the input images often possess exceedingly high feature non-linearity, segmenting the ROI specific regions become challenging, especially over overlapping objects of the same class. To address this problem, the proposed semantic segmentation model applied weighted loss concept, in which distinguishing background pixels or the labels between the overlapping cells retrieves a large weight in the loss function. The detailed discussion of the overall proposed UNet segmentation model and allied architecture (Fig. 1) is given as follows:

(a) UNet Architecture

As depicted in Fig. 1, the proposed UNet model comprises two paths, named contracting path and the expansive path, where the first is called left path, while the latter is known as the right path. In the deployed model, the contracting path was designed in its native structure depicting CNN encompassing two distinct 3×3 convolutions, where each convolution layer was

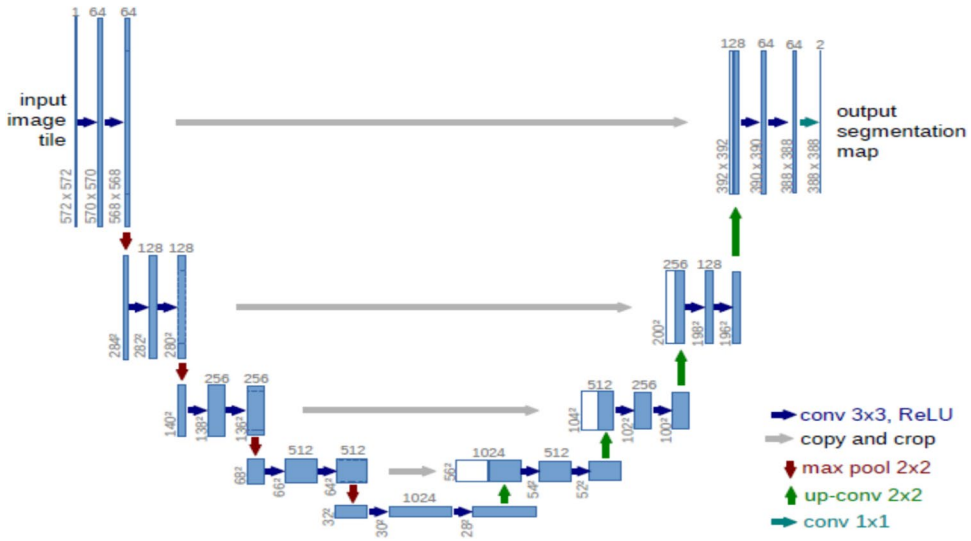


Fig. 1 Proposed UNet Architecture

applied in conjunction with the unit rectified linear unit (ReLU). Additionally, it also used 2×2 Max-Pooling layer designed with the stride size of two. Thus, the ReLU with the Max-pooling enabled down-sampling to retain high resolution features after each convolution. Each phase of the expansion path embodies feature's up-sampling task followed by 2×2 up-convolutions that eventually yields 50% ofthe total feature channels. Moreover, it applies a concatenation layer where the cropped feature map obtained from the contraction path gets fused to the two 3×3 convolutions, which is followed by unit ReLU unit. Noticeably, in at hand problem, cropping becomes inevitable due to the loss of the border pixels which often takes place in each convolution. Finally, at the final layer we used 1×1 convolution layer to map each 64-dimensional feature vector within the targeted number of classes. Thus, the overall design encompassed 23 convolutional layers that enabled seamless tiling of the resulting segmentation map.

(b) UNet Training for ROI Segmentation

To perform ROI segmentation, the proposed method applied the input images and corresponding segmentation maps to train the network. In this work, we applied ADAM non-linear learning method to perform training. Because of the unpadded convolutions, the output images were found smaller in size in comparison to the input images (here, it applies a fixed border width). To ensure optimal leverage between the computational costs and accuracy the batch size of 8 was taken into consideration. Though, reducing the batch sizes can help improving the accuracy, but at the cost of increased computation time. In the proposed training model, energy function was measured by means of a pixel-wise Softmax layer, applied over the final feature map amalgamated with the cross-entropy loss function. Here, we define Softmax layer as (6).

$$p_k(x) = \frac{exp\big(a_k(x)\big)}{\Big(\sum_{k'}^{k} exp\big(a_k(x)\big)\Big)} \tag{6}$$

In (6), the parameter $a_k(x)$ states the activation function for the k-th feature channel at the pixel position of $x \in \Omega$ with $\Omega \in Z^2$. The other parameters K represents the total number of classes, while the approximated minimum function is given by $p_k(x)$. In other words, $p_k(x) \approx 1$ for the k possessing the maximum activation $a_k(x)$, while for other values of k, it retains $p_k(x) \approx 0$ value. Now, it applies the cross-entropy to penalize at every position the deviation of $p_{l(x)}(x)$ from 1, by applying Eq. (7).

$$E = \sum_{x \in \Omega} w(x) \log\big(p_{l(x)}(x)\big) \tag{7}$$

In (7), the variable $l : \Omega \rightarrow \{1, \ldots, K\}$ be the true label of each comprising pixel, while $w : \Omega \rightarrow \mathbb{R}$ refers the weight map that we applied to provide some pixels higher significance (ROI) during training. The proposed method at first pre-estimates the weight map for each ground truth segmentation to compensate the disparity of the different pixel frequency within the training data. It enables learning of the the small distinction (say, separation) borders or edges to achieve higher accuracy. Noticeably, in the proposed method the separation border is measures by applying morphological functions, which is then followed by the estimation of the weight map, given as (8).

$$w(x) = w_c(x) + w_0.exp\left(-\frac{\big(d_1(x) + d_1(x)\big)^2}{2\sigma^2}\right) \tag{8}$$

In (8) $w_c : \Omega \rightarrow \mathbb{R}$ states the weight map applied to balanced the class-frequencies. The other component $d_1 : \Omega \rightarrow \mathbb{R}$ and $d_1 : \Omega \rightarrow \mathbb{R}$ represent the distance to the border of the closest cell and the distance to the border of the second nearest cell. In the deployed model $w_0 = 10$ and $\sigma = 5$ pixels. In the deep architecture alternating convolution and ReLU layers were applied, where for a 3×3 convolution encompassing 64 channels (in the descendent layer) a total number of incoming nodes of one neuron was estimated as $N = 9 * 64 = 576$. Thus, applying above discussed method, the ROI regions were segmented from each input image. The segmented outputs for the different input images are given in Table 2.

Realizing the fact that extracting deep feature require RGB images to extract more information intensive feature vector, we transformed the ROI specific regions into the ROI-specific RGB image. It was achieved by performing convolution between the ROI specific segmented region and the input original images. Some of the ROI-specific input images for further feature extraction are given in Table 3. In this manner, the obtained ROI specific RGB images were processed for feature extraction and learning, which is discussed in the subsequent sections.

(C) Deep Spatio-Textural Feature Extraction

Once obtaining the ROI specific RBG images, we executed the proposed deep spatio-textural feature extraction model by applying GLCM method and ResNet101 distinctly. Here, our motive was to exploit both GLCM (spatio-textural features) and

Table 2 ROI Segmented regions (Binary)

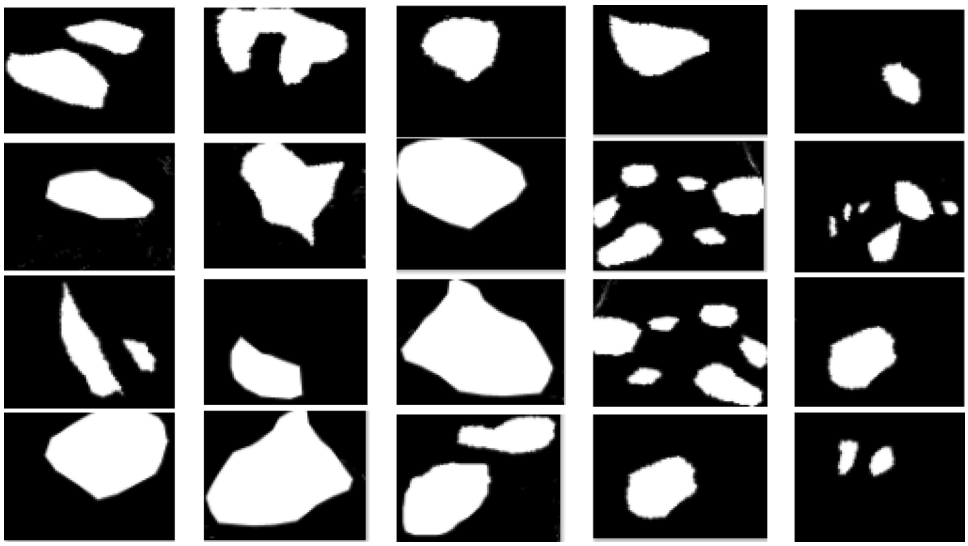


Table 3 Segmented ROI in RGB colour space



deep features (i.e., ResNet101) to achieve an information rich feature vector for superior training and hence higher leprosy detection and classification accuracy. The detailed discussion of the overall proposed feature extraction model is given as follows:

(a) Gray Level Co-Occurrence Matrix

In this work, a total of eight GLCM features representing spatio-textural, orientation related features were taken into consideration. A brief of these methods is given as follows:

(1) Contrast

In GLCM feature prospect, we define contrast as the variations in the Gray-scale values across the input ROI specific input images. Though to achieve it, at first, we estimated a matrix called probability matrix, given in Eq. (9). Based on the derived probability matrix, the other spatio-textural features were obtained as discussed below:

$$P_{i,j} = \frac{S_{i,j}}{\sum_{i,j=0}^{N-1} S_{i,j}} \tag{9}$$

Deriving the probability matrix (9), the pixel pairs representing the diagonal element yields the disparity in contrast values. Here, the textural contrast refers the cumulative changes in the local pixel intensities across the input ROI-specific leprosy image(s). In General, the non-linearity across the input image has been assessed by applying statistical assessment and corresponding textural continuity examination. We used Eq. (10) to measure contrast in an input image.

$$CONT = \sum_{i,j=0}^{N-1} P_{i,j}(i-j)^2 \tag{10}$$

(2) Energy

Here, energy distribution was measured across the input images (say, ROI specific RGB leprosy lesions). To measure energy distribution, at first, angular second moment (ASM) representing the rotational acceleration over the input feature space was measured as per (11). To be noted, typically ASM value increases linearly over increasing energy. Thus, estimating ASM per input image, the energy was calculated as per Eq. (12).

$$ASM = \sum_{i,j=0}^{N-1} P_{i,j}^2 \tag{11}$$

$$ENR = \sqrt{ASM_{i,j}} \tag{12}$$

(3) Homogeneity

Typically, homogeneity is measured in the form of the Inverse Different Moment (IDM) and therefore higher IDM signifies higher homogeneity. Homogeneity is also examined in terms of contrast, where lower contrast is hypothesized to be depicting higher homogeneity. In this work, we used Eq. (13) to measure homogeneity distribution across an input ROI-specific color lesion image.

$$HOM = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1+(i-j)^2} \tag{13}$$

(4) Correlation

In GLCM method, correlation signifies a descriptive statistical feature depicting the relatedness or associations amongst the different pixels. Employing the probability matrix value, as derived in $P_{i,j}$ (9), the different statistical features including mean (14–15) and standard deviation, were obtained for the different pixel distributed over the ROI spaces.

$$\mu_i = \sum_{i,j}^{N-1} i\left(P_{i,j}\right) \tag{14}$$

$$\mu_j = \sum_{i,j}^{N-1} j\left(P_{i,j}\right) \tag{15}$$

Subsequently, the measured values of the mean and were applied to measure standard deviation (17).

$$\sigma_i^2 = \sum_{i,j}^{N-1} P_{i,j}\left(i-\mu_i\right)^2 \tag{16}$$

$$\sigma_i = \sqrt{\sigma_i^2} \tag{17}$$

Thus, employing above derived statistical parameters, (i.e., mean and variance), the correlation of each input (ROI specific color) image was obtained as per the Eq. (18).

$$CORR = \sum_{i,j}^{N-1} P_{i,j}\left[\frac{\left(i-\mu_i\right)\left(j-\mu_j\right)}{\sqrt{\left(\sigma_i^2\right)\left(\sigma_j^2\right)}}\right] \tag{18}$$

(5) Skewness

In this work, we derived other orientation-related features, such as Skewness and Kurtosis by applying GLCM method (9). The targeted orientation features are also known as the symmetrical statistical features, where the Skewness of an input image refers the lack of symmetry across the input image. In this work, skewness is defined in terms of the shade feature in which the higher cluster-shade represents asymmetric nature of the feature space. Mathematically, skewness is measured for each input image by using (19).

$$SKEW = \sum_{i,j}^{N-1} P_{i,j}\left(i-\mu_i+j-\mu_j\right)^4 \tag{19}$$

(6) Kurtosis

Kurtosis (KURT) represents the 'peaked-ness' of an input gray-level values distributed throughput the ROI specific input (say, image). Typically, higher value of Kurtosis refers that the value of the feature-distribution is more strenuous towards tail(s) than the mean value. On the contrary, lower Kurtosis states that the magnitude of feature-distribution remains strenuous in the direction of the spike existing nearer to the mean value. In this work, Kurtosis was measured for complete ROI-specific input image.

Extracting above derived GLCM features, horizontal concatenation was performed to derive a structured (GLCM) feature vector by using concatenation function (20).

$$GLCM_{Feat} = Conc(CONT, ENE, HOM, CORR, Mean, STD, Kurt \tag{20}$$

(b) ResNet101 Deep Residual Feature Extraction

Unlike classical deep feature extraction model, such as CNN and RNNs, which are often criticized to undergo or suffer gradient vanishing problem, we applied ResNet as the deep feature model. Noticeably, the residual networks such as ResNet50 and ResNet101 are found effective to alleviate the issue of gradient fading that enables more intrinsic feature extraction towards leprosy feature learning and classification. In our work, we applied ResNet101 deep network, which is a modified version of ResNet50. Noticeably, the residual networks are often named based on the architecture it follows (say, total number of layers); yet, the root of ResNet101 can be found from VGG-19 deep network [18]. In general, a CNN embodies multiple layers which are connected to each other and are trained distinctly as well as in collaborative manner to perform learning and classification. Their network design enables learning of at varied layers and continues till the last layer of the network. Yet, unlike aforesaid deep networks, the convolutional layers in ResNet101 embodies almost 33 filters, helping it to retrieve more deep intrinsic features. In ResNet101, the layers possess similar or same number of filters for equivalent output

or the output feature map. Yet, the filter's count gets doubled if one wants to minimize feature map output (say, 50%) to minimize computational cost, time and complexity. To achieve it, ResNet101 employs down-sampling which is achieved by convolving layers with stride defined as 2. The proposed ResNet101 design terminates with a global average pooling layer and a SoftMax activated fully connected layer. The residual learning is achieved as the subtraction of input features which has been learnt from that specific layer. We achieved it by using a shortcut connection, also called shorting to every couple of the next filters (i.e., 33 distinct layered filters), by interfacing the input of $k-th$ layer to $(k+x)-th$ layer. The purpose of bypassing these layers is to address the problem of gradient vanishing problem which is achieved by re-employing activations from the subsequent layer till the layer next to the current one has learnt respective weights. For network training, learning weights amplifies the next layer to the current layer that enables training more efficient with fine grained intrinsic features and therefore enables it superior over the traditional deep network. The deployed ResNet101 alleviates the likelihood of the accuracy degradation, and hence makes deep network more accurate towards at hand leprosy detection. The ResNet101 network embodies a total of 101-layers to achieve learning.

For ResNet101 realization, the deployed deep network resized the pre-processed image and hence each image was resized to the dimension of $224 \times 224 \times 3$. Before feature extraction, the pixel values of each input ROI-specific image were normalized by applying (21).

$$\bar{x}(i,j) = \frac{x(i,j)}{127.5} - 1 \tag{21}$$

In (21), $x(i,j)$ states a vector containing pixel values with reference to the (i,j) cordinates. The deployed ResNet101 deep network performs multi-layer feature extraction, where the residual blocks were added with the previous layer to alleviate any probability of loss-function or accuracy degradation (sometimes, gradient vanishing). Noticeably, the aforesaid gradient vanishing and/or degradation issue can be highly frequent in image feature extraction due to the multiple non-linear layers in the input ROI-specific leprosy lesion images, which can impact performance in terms of the increased computation, complexity and delay. It can limit the scalability of such solution over real-time leprosy detection and classification system. To alleviate aforesaid problem and achieve swift convergence, the deployed ResNet101 deep network performed retraining of the inputs by adding a residual block, which can be defined as per the Eq. (22).

$$y = F\left(x, \{W_i\}\right) + x \tag{22}$$

In (22), x and y refers the input and allied output feature vectors for a given layer. The component $F\left(x, \{W_i\}\right)$ states the residual mapping information to be learnt. The residual mapping and residual bock addition function (22) (Fig. 2). Here, in the deployed ResNet101 the connected shortcut solves loss-function without burdening or applying additional parameters and hence intends to achieve better performance.

In ResNet101 deep network, a total of 101 layers were applied, where each layer contained 33 distinct filters to extract features and learn. Eventually, it applies dense layer connected with Softmax activation. Since, in our research we wanted to exploit both GLCM as well as ResNet101 features altogether, average pooling layer was applied followed by Softmax layer to retrieve ResNet101 feature vector (i.e., $Feat_{ResNet101}$).

D. Feature Fusion

Once extracting the GLCM spatio-textural features ($GLCM_{Feat}$) and ResNet101 features ($Feat_{ResNet101}$), horizontal concatenation was performed as per Eq. (23).

$$Feat = \left[GLCM_{Feat}; Feat_{ResNet101}\right] \tag{23}$$

The obtained composite feature vector (23) was the fed as input to the random forest ensemble algorithm to perform binary classification, which classifies each input image into two classes "Leprosy" or "Normal Lesion",

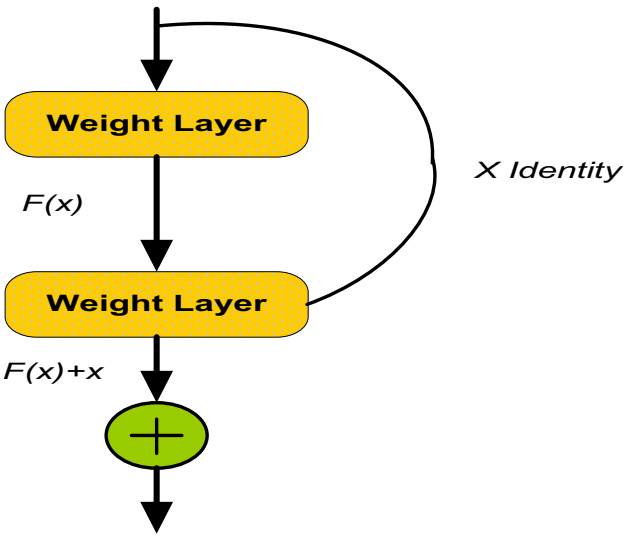


Fig. 2 Identity block

and annotates them as "1" and "0", respectively. Unlike classical machine learning methods, to improve classification reliability Random Forest (RF) ensemble algorithm was applied. Though, in this work, different machine learning algorithms were applied to perform feature learning and classification; however, considering space constraints only the best performing model (refer, Section V) RF algorithm is discussed. A brief of the RF ensemble learning method is given in the subsequent section.

E. Random Forest Ensemble Classification

Unlike traditional standalone machine learning classifiers, the RF classifier represent an ensemble learning method that embodies multiple tree-based classifiers, acting as an ensemble learning model to perform learning and classification. Being consensus-driven approach, RF classifier yields more reliable and generalizable performance than the classical machine learning methods. Moreover, the smaller number of parameters required in RF makes it superior over the other machine learning methods. Considering these features, we applied RF ensemble learning algorithm to perform learning and classification (for leprosy detection and classification). Being a tree-based model, each tree yields corresponding prediction result or vote for the probable class for each input image. Let, the overall input image counts be N. In this case, an image having N cases can be randomly selected which can subsequently be applied as training set to form a new tree. In case there exists M inputs, then the best split over m can be applied to split the node. Here, m was fixed during forest growth phase, where eventually it grows each tree to the largest extent. An RF algorithm can be defined as the amalgamation of the different tree-structures, as depicted in (24).

$$\left\{h\left(x,\theta_k\right), k = 1, 2, \ldots i \ldots\right\} \tag{24}$$

In (24), h states the classifier function, while $\left\{\theta_k\right\}$ indicates the random vector distributed identically. In the deployed RF model, each tree embodies unit vote for the most probable class for an input image x. The dimensionality of θ relies on its use in the tree formation. The predominant reason behind RF success is its ability to form each decision tree constituting the forest. The proposed method 70% data samples were applied as training set, while 30% were used as the out-of-bag data for further cross-validation during classification. In this manner, applying this method, the proposed RF algorithm learnt over the extracted features (23) and performed binary classification to label each input image as the "Leprosy Image" and the "Normal Image", and labels them as "1" and "0", correspondingly, which helped deriving the confusion matrix (CM) for further performance characterization. To be noted, in order to assess efficacy of the different machine learning algorithm towards at hand leprosy detection and classification system, we applied ANN, DT, SVM, k-NN and NB algorithms in addition to the RF ensemble method as discussed above. Here, the key motive was to assess the efficacy of the specific machine learning algorithm to yield superior performance towards at hand leprosy detection and classification model. Due to space constraints the detailed discussion of these machine learning classifiers is not given in this manuscript.

Results and Discussion

This research contributed a deep semantic segmentation driven region-of-interest sensitive deep spatio-textural feature learning framework for leprosy detection and classification. This work emphasized on improving multiple aspects including feature optimality as well as computation ecosystem. For instance, to ensure a cost-effective and scalable CAD solution, the proposed model was designed in such manner that it can perform leprosy detection and classification over the skin lesion images acquired by means of the normal camera setup. Here, the main motive was to provide an easy to use and cost-effective CAD solution towards leprosy detection and classification. In this reference, the acquired data samples representing both normal skin lesions as well as leprosy lesion images were processed for preprocessing. More specifically, the proposed model applied contrast adaptive intensity equalization, histogram equalization and Z-score normalization that alleviated or supressed major at hand image related adversaries to support more efficient feature extraction and learning. It also normalized pixel intensities and allied feature distribution to enable better feature extraction towards learning and classification. Considering limited data samples and allied class-imbalance probability, the proposed model at first employed UNet segmentation, a deep driven semantic segmentation approach which helped segmenting the ROI specific regions, while dropping background pixel elements. To ensure optimal segmentation accuracy over spatio-textural feature non-linearity UNet deep network was applied with ADAM non-linear optimization function where the initial learning rate was assigned as 0.0001. In addition, the total number of epochs were assigned as 200. The segmented ROI specific images were then multiplied with the colour space RGB images that resulted RGB-specific leprosy lesions possessing three channels presentation. Subsequently, towards deep spatio-textural feature presentation the proposed model applied GLCM and

ResNet101 deep network distinctly, where the earlier method generated a total of eight features including contrast, energy, homogeneity, correlation, mean, standard deviation, kurtosis and skewness. On the other hand, ResNet101 provided deep feature from the input ROI-specific leprosy lesions. To learn over aforesaid deep spatio-textural features (from GLCM and ResNet101 network) horizontal concatenation was performed that yielded composite feature vector for further learning and classification. To assess efficacy of the different machine learning algorithms towards at hand leprosy detection and classification model, we applied ANN, SVM, DT, NB, k-NN and RF classifiers. These algorithms performed binary classification to classify each input image as "Leprosy Detected" and "Normal Image", and labelled them as "1" and "0", respectively.

The proposed leprosy detection and classification model was designed by using MATLAB2021b software, while the simulation was done over the Central Processing unit configured with the Intel-i5 processor, operating at the frequency of 3.2 GHz and memory of 8 GB. To assess performance in this work, statistical performance parameters including leprosy prediction accuracy, precision, recall and F-score. To achieve these parameters, we obtained confusion metrics in the form of true positive (TP), true negative (TN), false positive (FP) and false negative (FN). A snippet of the statistical parameters used for performance characterization is given in Table 4.

Table 4 Performance parameters

Parameter	Mathematical expression
Accuracy	$\frac{(TN+TP)}{(TN+FN+FP+TP)}$
Precision	$\frac{TP}{(TP+FP)}$
Recall	$\frac{TP}{(TP+FN)}$
F-Score	$2.\frac{Recall.Precision}{Recall+Precision}$

(a) Intra-Model Assessment

Since, this research focused on improving both feature environment as well as computing elements, where the first method intended to exploit both GLCM as well as GLCM with ResNet101 features, we compared the relative efficacy by the different machine learning classifiers. Here, our key motive was to assess and identify the best performing model towards reliable and scalable leprosy detection and classification system. Table 5 presents the relative performance over GLCM features as well as ResNet101 feature by using the different machine learning classifiers.

Observing the results (Table 5), it can easily be found that the Naive Bayes (NB) algorithm with GLCM feature exhibits the accuracy and F-Measure of 93.20% and 0.88, respectively. On the contrary, the same machine learning model over hybrid feature encompassing GLCM and ResNet101 exhibits the accuracy and F-measure of 94.57% and 0.93. While assessing the efficacy with the k-NN classifier, we found that the algorithm (k-NN) exhibits the accuracy and F-measure of 92.51% and 0.94, respectively over GLCM feature. On the contrary, the same machine learning classifier over the proposed hybrid feature exhibits 95.03% accuracy and F-Measure of 0.938, which shows better accuracy over the GLCM as standalone feature. The simulation results also revealed that the SVM classifier yields accuracy and F-Measure of 91.96% and 0.88, respectively over GLCM feature. On the contrary, it exhibits accuracy and F-Measure of 94.28% and F-Measure of 0.92. The decision tree (DT) algorithm as standalone classifier exhibited accuracy and F-Measure of 95.52% and 0.93, respectively. The simulation results also indicate that the same machine learning classifier exhibited accuracy and F-Measure of 95.45% and 0.93 over the proposed hybrid feature. It indicates that the DT machine learning method retains similar performance

Table 5 Intra-model performance assessment

Features/Model	Machine Learning	Accuracy (%)	Precision	Recall	F-Measure
GLCM	NB	93.20	0.91	0.98	0.88
	k-NN	92.51	0.91	0.99	0.94
	SVM	91.96	0.73	0.99	0.88
	DT	95.52	0.84	0.88	0.93
	ANN	95.59	0.93	0.98	0.94
	RF	95.63	0.93	0.98	0.95
Hybrid Feature (GLCM + ResNet101)	NB	94.57	0.91	0.98	0.93
	k-NN	95.03	0.87	0.99	0.93
	SVM	94.28	0.91	0.93	0.92
	DT	95.45	0.93	0.98	0.93
	ANN	95.19	0.91	0.99	0.96
	RF	97.24	99.69	95.82	0.98

for both feature models. Though, its homogenous ensemble, also called RF ensemble can be hypothesized to yield superior and more reliable performance. In this reference, the simulation results confirmed that the RF ensemble learning method in sync with GLCM feature exhibited the accuracy of 95.63% and F-Measure of 0.95. On the contrary, the RF algorithm with the proposed hybrid feature exhibited the accuracy of 97.24% and F-Measure of 0.98. The ANN method, which was applied with the Levenberg Marquardt learning method exhibited the accuracy and F-Measure of 95.2% and 0.96, respectively. Thus, observing overall results and allied inferences, it can be stated that the proposed RF ensemble method exhibits superior efficacy than the other machine learning methods towards at hand leprosy detection and classification system. Here, two inferences can be found, first that the use of the proposed UNet segmentation assisted hybrid feature model encompassing GLCM and ResNet101 feature can yield the optimal and the best possible performance towards leprosy detection and classification. In comparison to the GLCM standalone feature driven solution, the use of GLCM with ResNet101 residual features can perform superior towards the proposed leprosy detection and classification system. Consequently, it confirms affirmative answer for the research questions RQ1 and RQ2. Similarly, the simulation results (Table 5) confirms that the use of RF ensemble learning method can perform superior over the other machine learning methods and therefore confirms positive answer for the RQ3. Thus, the overall realization and allied performance characterization confirm that the proposed UNet semantic segmentation assisted deep spatio textural features embodying GLCM and ResNet101 feature vector can perform superior and optimal with the RF ensemble classifier. Thus, cumulatively it confirms the positive outcome for the research question RQ4.

(b) Inter-Model Assessment

Undeniably, merely a very few efforts have been made towards vision-based leprosy detection and classification task; however, based on the critical review, a few approaches have been identified to perform relative performance analysis. In inter-model assessment, we focus to assess whether the proposed model yields superior over the other state-of-art leprosy detection and classification systems. We identified a few state-of-art approaches applying local binary patterns [16], GLCM [16], and convolutional neural network (CNN) for leprosy image classification [27]. To be noted, the depth assessment reveal that the almost all existing approaches directly feed leprosy lesion images containing both ROI as well as non-ROI space to perform feature extraction and learning. In these approaches, the likelihood of biased learning (as the presence of ROI space can be quite smaller in comparison to the non-ROI spaces across 2D image(s)) can't be ignored and hence such approaches might give false positive performance. Unlike these approaches, in this work UNet assisted ROI-segmentation was performed where only ROI-specific regions were retained for further feature extraction. It helped alleviating or reducing irrelevant training and hence can be hypothesized to have impact on better performance. On the other hand, a recent work by Baweja et al. [27] who applied CNN for leprosy detection and classification considered whole input image as feature space to train the model. Their work neither addressed the problem of class-imbalance nor the gradient vanishing which is often incurred due to the drop of features over multiple convolutional layers. It can make their performance skewed or inaccurate. Unlike these state-of-arts, the proposed model used UNet driven segmentation which helped alleviating the class-imbalance problem, while the use of ResNet101 with each convolutional layer armoured with 33 filters alleviated the likelihood of gradient vanishing problems. It hypothesises the superior efficacy of the proposed model towards at hand leprosy detection and classification. To assess relative performance, we compared the efficacy of the proposed model with the other existing state-of-arts method. The relative performance of the different approaches is given in Table 6.

Baweja et al. [27] who applied CNN method over leprosy images to perform leprosy prediction. Despite the effort to improve CNN with Google inception-v3 model for leprosy region segmentations and learning, the highest accuracy observed was 93%, which is significantly lower than our proposed method (97.24%). A similar effort was made by Jain et al. [28] who applied CNN towards leprosy image classification. Unfortunately, with the classical pre-processing methods like median filtering, authors could merely achieve the average accuracy of 91% which is significantly lower than our proposed model (97.24%). Mondal et al. [29] too applied a modified deep learning architecture named Generative Adversarial Network (GAN) which was further improved with DensNet-121 could achieved the highest leprosy image classification accuracy of 94.25%. However, authors found that this claimed performance (i.e., the accuracy of 94.25%) is possible only with data augmentation. Interestingly, without data augmentation, the same GAN model exhibited accuracy of almost 83%, which is much lower than our proposed model. The comparison with other deep learning models like MobileNet V2 (81.6%), ResNet-101 (87.35%)

Table 6 Inter-Model Performance Assessment

References	Features	Accuracy (%)	Precision (%)	Recall (%)	F-Score
Lo et. al. [30]	DCNN	73.8	–	93.5	–
Mondal et al. [29]	GAN + DenseNet-121	94.25	94.0	90.4	0.922
	MobileNet V2	81.6	–	–	–
	ResNet-101	87.35	–	–	–
	GoogLeNet	89.65	–	–	–
	DenseNet-169	93.67	–	–	–
	DenseNet-121	94.25	–	–	–
Das et al. [16]	LBP	89.65	–	–	
	GLCM	88.51			
	DCI	73.56			
	DFT	70.69			
Jain et a. [28]	CNN	91.0	–	–	–
Baweja et. al. [27]	CNN + Inspection-v3	93.0	–	–	–
Proposed Method	GLCM + ResNet101	97.24	99.69	95.82	0.98

and GoogleNet (89.65%) too are found lower than the proposed hybrid features driven solution. Noticeably, despite the use of ResNet101 in [29] the authors could get the highest accuracy of 87.35%, which was only because the authors fed entire input image for feature extraction and learning. Moreover, they applied cross-entropy feature learning to perform binary classification. Unlike their work, we at first performed ROI specific semantic segmentation followed by hybrid deep spatio-textural feature extraction and learning, which yielded superior accuracy of 97.24% than the state of art [29]. It clearly confirms that the amalgamation of deep spatio-textural features (say, GLCM + ResNet101) enabled the proposed model yielding superior accuracy towards at hand leprosy detection and classification system design. It confirms acceptance or affirmative answer for the RQ1-RQ4.

Lo et al. [30] too applied deep-CNN model (DCNN) in conjunction with AlexNet to perform bacillus image blocks classification. In addition to the fact that the images considered were not in sync with real-world diverse leprosy lesions and allied non-linear spatio-temporal textures, the highest accuracy obtained was 73.8%. In this approach, authors applied a static threshold value of 0.5, where each image block was assessed for DCNN feature values. An image with probability value more than 0.5 was classified as bacillus. In sync with real-time non-linear and diverse leprosy lesions, this approach can't be guaranteed to exhibit reliable performance, especially over varying skin colours, non-linear shapes, sizes and orientation (due to the lesion presence on the different body parts). Noticeably, the proposed model addresses aforesaid real-time challenges and its functional efficacy confirms its robustness over the other existing approaches in terms of higher accuracy (97.24%%), precision (99.69%), recall (95.82%) and F-measure of 0.98. The F-Measure value signifying the reliability of the classifier over non-linear and minimally annotated data condition too has been found sufficient (0.98), thus guaranteeing the robustness of the proposed model towards touchless, automatic vision-based leprosy detection and classification system. Considering the contribution by Das et al. [16] who explored the different approaches including local binary patterns (LBP), GLCM, Discrete Cosine Transform (DCT), Discrete Fourier Transform (DFT) towards leprosy image classification found that the DCT-features can yield the highest accuracy of 73.56%, while the highest accuracy performance with GLCM was observed as 88.51%. In the same research, LBP was found exhibiting the leprosy image classification accuracy of 89.66%. DFT-based approach yields the least performance with the classification accuracy of 70.7%. The graphical depiction of the relative performance by the different methods including our proposed hybrid feature driven model (*GLCM + ResNet101) is given in the following figure. The comparison results clearly indicate that the proposed hybrid feature driven model (i.e., *GLCM + ResNet101) outperforms other existing approaches or other feature driven model for vision-based leprosy prediction [47].

Conclusion

In sync with vision computing-based leprosy detection and classification system, this research contributed a robust deep semantic segmentation assisted region-of-interest sensitive deep spatio-textural feature learning framework for leprosy detection and classification. To achieve it, the proposed

method focused on multiple aspects including ROI-specific learning, feature optimization by amalgamating deep spatio-textural features, the use of residual network to alleviate gradient vanishing problems and eventually ensemble-based learning for reliability. Moreover, to ensure cost-efficient and scalable solution, the proposed model applied normal camera-acquisition based inputs (images), which were at first processed for pre-processing by performing contrast adaptive histogram equalization, intensity equalization and z-score normalization. It helped alleviating the impact of local factors on further processing. Subsequently, deep driven semantic segmentation was performed over the input images which performed pixel-wise classification to detect the ROI by dropping the background elements. It helped retaining the ROI for further ROI-specific feature extraction and learning and hence alleviated any likelihood of class-imbalance. Additionally, it enabled ROI-specific feature extraction which can be significant towards learning and classification. The segmented ROI regions were then processed for RGB multiplication so as to retain ROI specific RGB images for further deep spatio-textural feature extraction. Unlike classical approaches, where the authors have merely applied GLCM, LBP or CNN as standalone feature extraction model, the proposed model employed GLCM along with the residual network ResNet101. More specifically, seven GLCM features characterising the spatial feature distribution and textural details including entropy, energy, homogeneity, correlation, mean, standard deviation and variance were obtained. On the other hand, the use of ResNet101 provided information rich feature vector by alleviating any probability of gradient vanishing issue. Thus, the obtained GLCM (spatio-textural) and ResNet101 features were processed for feature-level fusion and thus the fused feature vector was applied as input to the random forest classifier for leprosy detection and allied classification tasks. The simulation over the primary collected leprosy datasets exhibited accuracy and F-Measure of 97.24% and 98%, respectively. The performance outcomes confirm efficacy of the proposed model to serve leprosy detection for real-time and scalable CAD demands.

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Declarations

Conflict of interest No conflict of interest.

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